

Stephen Tsai

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Education

Virginia Polytechnic Institute and State University

Bachelor of Science in Computer Science: GPA: 3.79

Blacksburg, VA

Expected Graduation: May 2028

Relevant Coursework:

Linear Algebra | Software Design & Data Structures | Discrete Mathematics | Computer Organization I |

Experience

Virginia Tech College of Engineering - Blacksburg, VA

Jan 2025 - June 2025

Undergraduate Researcher - Machine Learning and Natural Language Processing (NLP)

- Built Python-based NLP pipelines to collect and analyze 500+ university AI policy webpages and 1,000+ OCR-converted PDFs
- Evaluated and benchmarked 5+ text-processing and LLM pipelines, improving large-scale policy analysis throughput by ~30%
- Extracted structured policy metadata using HuggingFace models and custom preprocessing for nationwide trend analysis
- Funded research by the Institute for Society, Culture and Environment

Alarm.com - Tysons, VA.

June 2022 - Aug 2022

Administration / IT Intern

- Coordinated with IT and administrative teams to resolve internal technical support requests, improving operational efficiency
- Assisted with internal systems, tooling support, and technical documentation while gaining exposure to production engineering workflows
- Gained exposure to production software engineering workflows through cross-team collaboration and technical seminars

Projects

Simplify | Python, Flet, Ollama (LLaMA 3.1)

Dec 2025 - Present

- Built a cross-platform desktop application that summarizes and simplifies highlighted text from any application using global hotkeys and clipboard integration
- Integrated a locally hosted large language model (Llama 3.1) with Python to perform simplification and summarization of text
- Designed a custom and responsive GUI using Flet to display real-time input and processed output

ARM (Automated Recyclable Management) - [Article](#) / [Abstract](#)

- Designed and trained an object detection model to classify recyclable materials across multiple waste categories
- Built and integrated a custom hardware prototype to automate waste sorting using real-time computer vision inference
- Performed dataset collection, labeling, and model evaluation to improve detection accuracy under varying lighting and positioning conditions
- Deployed and demonstrated an end-to-end vision to hardware pipeline through live testing and competition presentations

EyeSpy - [GitHub](#)

- Created a real-time driver drowsiness detection system utilizing computer vision techniques to enhance road safety
- Integrated TensorFlow pipeline with audio alerts deploying Pygame for eye closure detection

ShooterStopper - [GitHub](#)

- Developed an AI-powered surveillance system to identify firearms in real-time surveillance footage
- Deployed EfficientDet via Flask REST API and integrated with React.js frontend for real-time video monitoring
- Logged detection events to CockroachDB for real-time threat history tracking

Involvements & Skills

Awards: Regeneron ISEF Finalist (Top 1,800 students across 75+ countries) | Winner at Poolesville_hacks and OneHacksII | Dean's List with Distinction (Awarded to students with a GPA higher than 3.75) | Virginia Tech Table Tennis Team

Languages: Python, Java, JavaScript, Bash, C,

Frameworks / Tools: Linux, TensorFlow, HuggingFace, GitHub, React, OpenCV

Libraries: Pandas, NumPy, Matplotlib, Scikit-learn